Inequalities: When Does the Symbol Flip?

The one rule this page is about. Multiplying or dividing *both sides* of an inequality by a *negative* number reverses the inequality symbol. Adding or subtracting a negative number does *not*. Every problem below is a place where that rule is either used or forgotten.

Directions. Show each step. When you divide or multiply by a negative, write "flip" beside that step so a reader can see where the symbol turned around. For the find-and-fix problems, name the error in words before you give the correct answer.

1. The statement $6 > -2$ is true. Multiply *both* sides by -4.

Left side: $-4 \times 6 = -24$. Right side: $-4 \times (-2) =$ __________

Now compare your two results and circle the symbol that makes the new statement true: -24 $\boxed{>}$ $\boxed{<}$ (your answer). What did the original ">" become?

Answer: __________________

2. Solve for x: $-4x > 12$

Write the solution as an inequality, and state which step forced the flip.

Answer: __________________

3. **Find and fix.** Priya solved $-7x \leq 21$ like this:

$$-7x \leq 21 \;\longrightarrow\; \frac{-7x}{-7} \leq \frac{21}{-7} \;\longrightarrow\; x \leq -3$$

Her arithmetic is correct, but her answer is wrong. **(a)** Test $x = -10$ in the *original* inequality to show her answer cannot be right. **(b)** Name her error in one sentence. **(c)** Write the correct solution.

Answer: ________________

4. Solve for x: $-\dfrac{x}{3} < 2$

Dividing by 3 and multiplying by -1 are two different moves — only one of them flips the symbol.

Answer: ________________

5. Solving $-5x - 2 \leq 18$ gives the boundary value $x = -4$. Check that boundary: substitute $x = -4$ into the expression $-5x - 2$ and evaluate it.

If your result equals 18, the boundary is correct. (A student who gets -22 here has broken the negative-times-negative rule — check your signs.)

Answer: ________________

6. Solve for x: $4 - 3x > 19$

Undo the $+4$ first. Only then do you meet the negative coefficient.

Answer: ________________

7. Find and fix. Dev solved $8 - 2x \geq 14$ like this:

$$8 - 2x \geq 14 \longrightarrow -2x \geq 6 \longrightarrow x \geq -3$$

(a) Which of Dev's two steps is wrong, and why is the other one fine? **(b)** Show that $x = 0$ satisfies Dev's answer but not the original inequality. **(c)** Write the correct solution.

Answer: ________________

8. Explain. Adding -5 to both sides of an inequality never changes the symbol, but multiplying both sides by -5 always does. Using a number line (or a pair of specific numbers of your own) explain why those two moves behave differently. Your explanation should say what multiplying by a negative does to the *positions* of two numbers on the line.